

Year 11 Term 4 Mathematics Extension 1, 2008

Question 1

Marks

(a) Solve for x : $\frac{3x+2}{2x} \geq 4$

3

(b) Find the general solution to : $1 + 2\sin x = 0$

2

(c) Differentiate with respect to x and simplify :

(i) $2 \tan^{-1}\left(\frac{3x}{4}\right)$

3

(ii) $\cos^{-1}(2x-3) + \sin^{-1}(2x-3)$

1

Question 2

(a) Evaluate $\int_1^2 x\sqrt{x^2+2} dx$ using the substitution $u = x^2 + 2$.

3

(b) Find $\int \frac{4dx}{\sqrt{25-4x^2}}$

2

(c) (i) Express $2\sqrt{3}\sin x - 2\cos x$ in the form $R\cos(x+\alpha)$ where $R > 0$
and $-\frac{\pi}{2} \leq x \leq \pi$.

2

(ii) Hence or otherwise solve : $2\sqrt{3}\sin x - 2\cos x > 2$ for
 $-\frac{\pi}{2} \leq x \leq \pi$.

2

Question 3

(a) (i) Find $f'(x)$ if $f(x) = \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right)$.

2

(ii) Deduce an equivalent function for $f(x)$.

1

(iii) Hence graph the function $y = \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right)$.

2

(b) (i) Graph the curve $y = 3\cos^{-1} x$.

2

(ii) Find the exact area of the region bounded by the curve
 $y = 3\cos^{-1} x$, the lines
 $y = \pi$ and the y axis.

2

Question 4

- (a) (i) On the same axes shade the region defined by : 3
 $y \leq 2x + 1$, $y \geq x^2 - 2$ and $x \geq 2$.

- (ii) Show that the point $P(3, 7)$ is an intersection point of $y = 2x + 1$ 2
and $y = x^2 - 2$.

- (iii) Find the volume of revolution when the region defined in (a)(i) is 4
rotated about
the x axis.

Question 5

- (a) If $A = \cos^{-1}\left(\frac{8}{17}\right)$ find the value of $\tan 2A$. 2

- (b) Evaluate $\int_2^{2\sqrt{3}} \frac{x-2}{\sqrt{16-x^2}} dx$ 4

- (c) If $f(x) = 2^x + 5$ find the inverse function $f^{-1}(x)$, stating the 3
domain and range of $f^{-1}(x)$.

Question 6

- (a) Evaluate to 3 decimal places $\int_0^4 \log_{10}(x^2 + 3) dx$ using Simpson's Rule 3
with 3 function values.

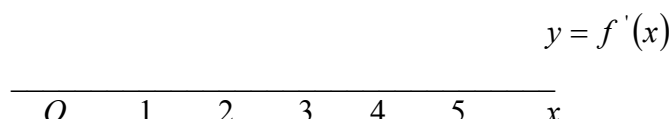
- (b) (i) Prove $\frac{x+y}{2} \geq \sqrt{xy}$ for all $x \geq 0$ and $y \geq 0$. 2

- (ii) Hence prove that the area of a rectangle cannot exceed the average 1
of the square of
the lengths of its two sides.

- (iii) If $a > 0$, $b > 0$, $c > 0$, and $a + b + c = 1$, show that 3
 $(1-a)(1-b)(1-c) \geq 8abc$.

Question 7

- (a) The graph below shows the gradient function $f'(x)$. 2
 y



Find the values of x where $y = f(x)$ has a local minimum. Justify your answer.

- (b) A truck is to travel 1000 kilometres at a constant speed of v km/h.
When travelling at v km/hr, the truck consumes fuel at the rate of

$$\left(6 + \frac{v^2}{50} \right) \text{ litres per hour.}$$

The truck company pays 50 cents/litre for fuel and pays each of the two drivers

\$20 per hour whilst the truck is travelling.

2

- (i) Let the total cost of fuel and the drivers' wages for the trip be C dollars. Show that :

$$C = 10v + 43000 \frac{1}{v}.$$

- (ii) The truck must take no longer than 12 hours to complete the trip, and speed limits

5

require that $v \leq 100$.

At what speed v should the truck travel to minimize the cost C ?

End of Exam

Q1 SU Y4 T4.

(a) $\frac{3x+2}{2x} \geq 4 \quad x \neq 0.$

$$2x(3x+2) \geq 16x^2$$

$$x[3x+2-8x] \geq 0.$$

$$x[2-5x] \geq 0$$

$$0 < x \leq \frac{2}{5}$$

(3)

(b) $\sin x = -\frac{1}{2}$

$$x = \pi k + (-1)^k \sin^{-1}\left(-\frac{1}{2}\right)$$

$$= \pi k - (-1)^k \frac{\pi}{6} \quad k \text{ integer. } (2)$$

(c) (i) $\frac{d}{dx} 2 \tan^{-1}\left(\frac{3x}{4}\right) = 2 \cdot \frac{3}{4} \cdot \frac{1}{1 + \left(\frac{3x}{4}\right)^2}$

$$= \frac{3}{2} \cdot \frac{1}{1 + \frac{9x^2}{16}}$$

$$= \frac{24}{16 + 9x^2} \quad (3)$$

(ii) $\frac{d}{dx} \left[\cos^{-1}(2x-3) + \sin^{-1}(2x-3) \right] = \frac{d}{dx} \left(\frac{\pi}{2} \right)$

$$= 0. \quad (1)$$

YR11 T4 2008 EXT 1

Q2(a) $\int_1^2 x \sqrt{x^2+2} \, dx$

$$\begin{aligned} u &= x^2+2 & x=1 & u=3 \\ du &= 2x \, dx & x=2 & u=6 \end{aligned}$$

$$= \int_3^6 u^{\frac{1}{2}} \frac{du}{2}$$

$$= \left[\frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right]_3^6$$

$$= 2\sqrt{6} - \sqrt{3} \quad (3)$$

(b) $\int \frac{4 \, dx}{\sqrt{25-4x^2}} = \frac{4}{2} \int \frac{dx}{\sqrt{\frac{25}{4} - x^2}}$

$$= 2 \sin^{-1}\left(\frac{x}{\frac{5}{2}}\right) + C$$

$$= 2 \sin^{-1}\left(\frac{2x}{5}\right) + C \quad (2)$$

c(i) $2\sqrt{3} \sin x - 2 \cos x = R \cos(x+d) \quad R > 0.$

$$= R \cos x \cos d - R \sin x \sin d$$

$$R \cos d = -2\sqrt{3} \quad \sin d > 0$$

$$R \sin d = -2 \quad \cos d > 0$$

$$\tan d = \sqrt{3}$$

$$d = \frac{4\pi}{3}$$

$$R = \sqrt{(2\sqrt{3})^2 + 2^2}$$

$$= 4$$

$$2\sqrt{3} \sin x - 2 \cos x = 4 \cos\left(x + \frac{4\pi}{3}\right)$$

(ii)

$$4 \cos\left(x + \frac{\pi}{3}\right) = 2$$

$$\cos\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$x + \frac{\pi}{3} = 2\pi k \pm \frac{\pi}{3}$$

$$x = \pi \text{ or } \frac{\pi}{3}$$

$$\therefore \frac{\pi}{3} < x < \pi$$

Q.3 (i) $f(x) = \tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)$

$$f'(x) = \frac{1}{1+x^2} + \frac{1}{1+\left(\frac{1}{x}\right)^2} \cdot \frac{-1}{x^2}$$

$$= \frac{1}{1+x^2} \cdot \frac{-1}{x^2\left(1+\frac{1}{x^2}\right)}$$

$$= \frac{1}{1+x^2} \cdot \frac{-1}{1+x^2}$$

$$= 0$$

(ii) (i) $f(x) = \text{constant}$

$$x > 0 \quad x=1 \quad f(x) = \tan^{-1}1 + \tan^{-1}1$$

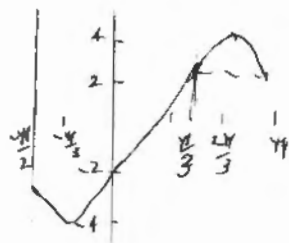
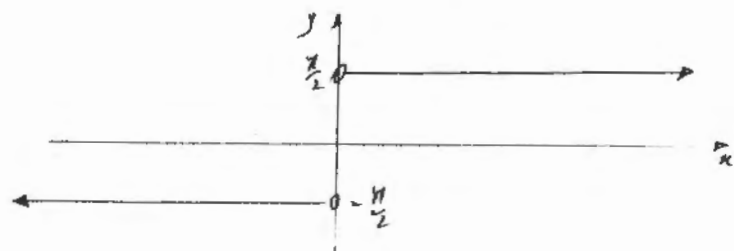
$$= \frac{\pi}{4} + \frac{\pi}{4}$$

$$x < 0 \quad x=-1$$

$$f(x) = -\frac{\pi}{4} - \frac{\pi}{4}$$

$$= -\frac{\pi}{2}$$

(ii)



(2)

3. (i) $y = 3 \cos \frac{x}{3}$

(ii) $x = \cos \frac{y}{3}$

$$\text{Area} = \int_{\pi}^{\frac{3\pi}{2}} \cos\left(\frac{y}{3}\right) dy$$

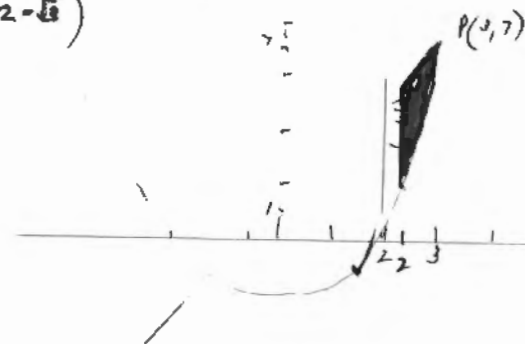
$$= \left[3 \sin \frac{y}{3} \right]_{\pi}^{\frac{3\pi}{2}}$$

$$= 3 \left[1 - \frac{\sqrt{3}}{2} \right]$$

$$= \frac{3}{2} (2 - \sqrt{3})$$

(2)

Q.4 (i)



(3)

(ii) $y = 2x+1$

$$x=3 \quad y=7$$

$$P(3, 7)$$

$$y = x^2 - 2$$

$$x=3 \quad y=3^2-2$$

$$y=7$$

satisfies both $y = 2x+1$ & $y = x^2-2$
 $\therefore P(3, 7)$ is an intersection point.

(iii)

$$\text{Volume } V = \pi \int_2^3 (2x+1)^2 - (x^2-2)^2 dx$$

$$= \pi \left\{ \left[\frac{(2x+1)^3}{6} \right]_2^3 - \int_2^3 (x^4 - 4x^2 + 4) dx \right\}$$

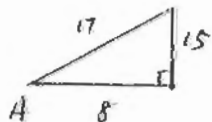
(4)

(1)

(1)

$$\begin{aligned}
 V &= \pi \left\{ \frac{7^3 - 5^3}{6} - \left(\frac{x^3}{5} - \frac{4x^3}{3} + 4x \right)_2 \right\} \\
 &= \pi \left\{ \frac{104}{3} - \left(\frac{243}{5} - 36 + 12 - \left(\frac{32}{5} - \frac{32}{3} + 8 \right) \right) \right\} \\
 &= \frac{232\pi}{15} \text{ unit}^3
 \end{aligned}$$

Q. 5



$$0 < A < \frac{\pi}{2}$$

$$\begin{aligned}
 \tan 2A &= \frac{2 \tan A}{1 - \tan^2 A} \\
 &= \frac{2 \cdot \frac{15}{8}}{1 - \frac{225}{64}} \\
 &= \frac{240}{64 - 225} \\
 &= -\frac{240}{161} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 (b) \int_2^{2\sqrt{3}} \frac{x-2}{\sqrt{16-x^2}} dx &= \int_2^{2\sqrt{3}} \left(\frac{x}{\sqrt{16-x^2}} - \frac{2}{\sqrt{16-x^2}} \right) dx \\
 &= \left[-\sqrt{16-x^2} - 2 \sin^{-1} \frac{x}{4} \right]_2^{2\sqrt{3}} \\
 &= -2 - 2 \cdot \frac{\pi}{3} - \left(-\sqrt{12} - 2 \cdot \frac{\pi}{6} \right) \\
 &= 2\sqrt{3} - 2 - \frac{\pi}{3}
 \end{aligned}$$

(1)

(1)

(2)

(2)

$$(c) \quad x = 2^y + 5 \quad x > 5$$

$$2^y = x - 5$$

$$y = \log_2(x-5)$$

$$f^{-1}x = \log_2(x-5) \quad (1)$$

Domain $f^{-1}x \quad x > 5$

Range $f^{-1}x$ defined all values of x

Ans

x	0	2	4
y	$\log_{10} 3$	$\log_{10} 7$	$\log_{10} 19$

$$\begin{aligned}
 3) \quad \int_0^4 \log_{10}(x^2+3) dx &\approx \frac{2}{3} \left[\log_{10} 3 + 4 \log_{10} 7 + \log_{10} 19 \right] \\
 &\approx \frac{2}{3} \log_{10} 136857 \\
 &\approx 3.424 \quad (3)
 \end{aligned}$$

$$b) (i) (\sqrt{x} - \sqrt{y})^2 \geq 0$$

$$\therefore x - 2\sqrt{x}\sqrt{y} + y \geq 0$$

$$x + y \geq 2\sqrt{xy}$$

$$\frac{x+y}{2} \geq \sqrt{xy} \quad (2)$$

$$(ii) \quad x = a^2 \quad y = b^2$$

$$\frac{a^2 + b^2}{2} \geq \sqrt{a^2 b^2}$$

$$\begin{aligned}
 a^2 + b^2 &\geq 2ab \\
 &\geq 2A
 \end{aligned}$$

$$\therefore A \leq \frac{a^2 + b^2}{2} \quad (1)$$

(1)

$$\begin{aligned}
 \text{iii) } & (1-a)(1-b)(1-c) \\
 & = (b+c)(a+c)(a+b) \quad \textcircled{1} \\
 & \geq 2\sqrt{bc} \cdot 2\sqrt{ac} \cdot 2\sqrt{ab} \quad \textcircled{1} \\
 & \geq 8\sqrt{a^2 b^2 c^2} \quad \textcircled{1} \\
 & \geq 8abc. \quad \textcircled{3}
 \end{aligned}$$

Q7(a) For minimum $f'_x = 0$
 $f''_x > 0$ (slope of f'_x at x)
 $\Rightarrow x = 5$ ②

(i) Cost = Cost driver + Cost fuel $t = \frac{1000}{v}$ km

$$\begin{aligned}
 & = \$2 \times 20 \times \frac{1000}{v} + \frac{1000}{v} \left(6 + \frac{v^2}{50} \right) \times \$0.50 \quad \textcircled{1} \\
 & = \$ \left(\frac{40000}{v} + \left(\frac{6000}{v} + 20v \right) \times 0.50 \right) \\
 & = \$ \left(\frac{40000}{v} + \frac{3000}{v} + 10v \right)
 \end{aligned}$$

$C = \$ \left(\frac{43000}{v} + 10v \right)$ ①

(ii) For minimum cost $\frac{dC}{dv} = -\frac{43000}{v^2} + 10$ ①

$$\frac{d^2C}{dv^2} = \frac{86000}{v^3}$$

For minimum cost $\frac{dC}{dv} = 0$.

$$-\frac{43000}{v^2} + 10 = 0$$

$$v^2 = 4300$$

$$v = 65.57 \text{ km/hr.} \quad \textcircled{1}$$

Time journey = 15.25 hrs.

Domain of v $t \leq 12 \text{ hrs}$ $v \leq 100$

$$\therefore 83\frac{1}{3} \leq v \leq 100. \quad \textcircled{1}$$

Since there is no turning point in domain
 then minimum cost is at the endpoint of domain ①

$$v = 83\frac{1}{3} \quad C = 1344$$

$$v \leq 100 \quad C = 1430$$

$\therefore$ Minimum cost when $v = 83\frac{1}{3} \text{ km/hr.}$ ①